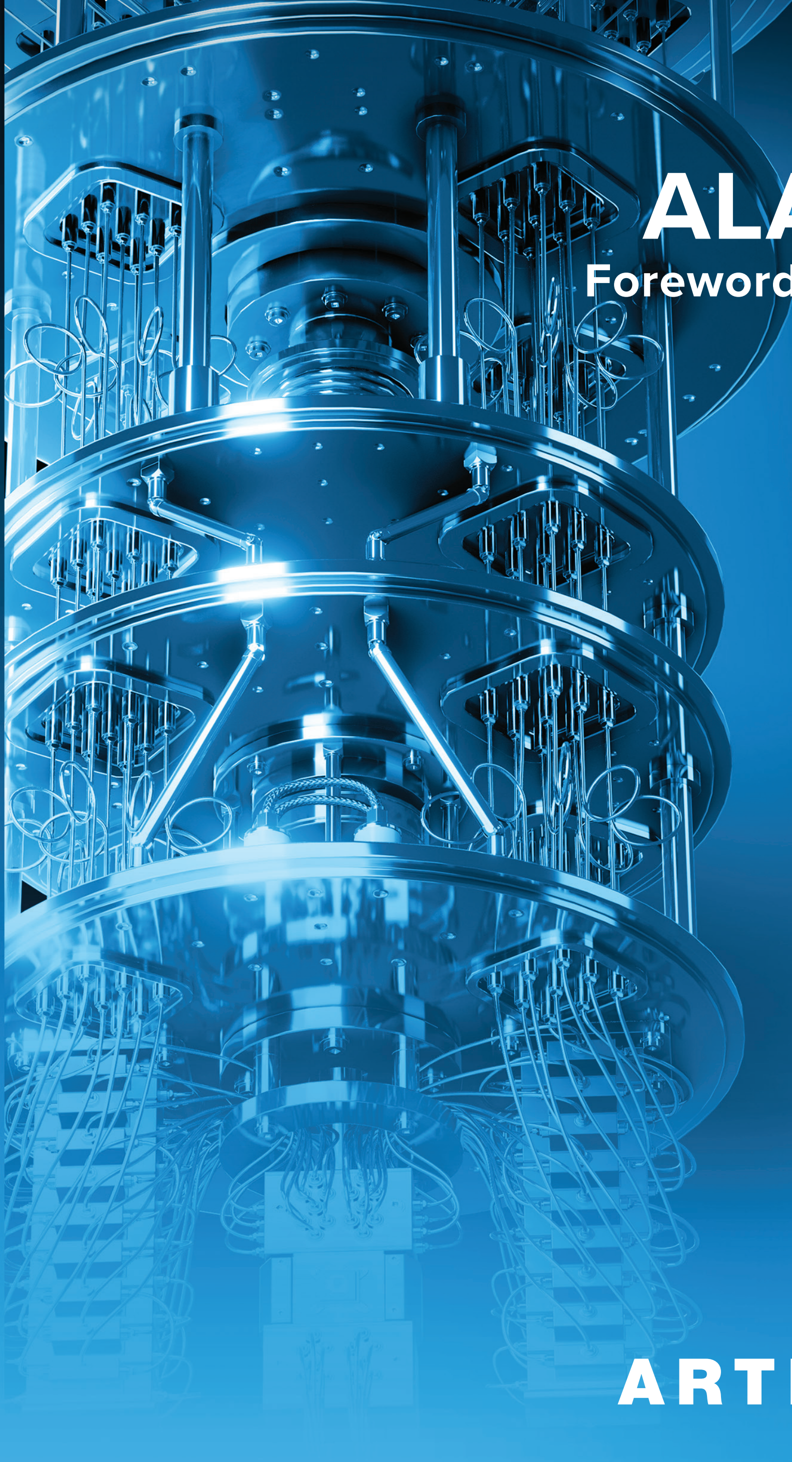
MICROWAVE TECHNIQUES IN SUPERCONDUCTING QUANTUM COMPUTERS



ALAN SALARI
Foreword by David DiVincenzo

ARTECH BOOKS

Microwave Techniques in Superconducting Quantum Computers

For a complete listing of titles in the
Artech House Microwave Library,
please turn to the back of this book.

Microwave Techniques in Superconducting Quantum Computers

Alan Salari

ARTECH HOUSE
BOSTON | LONDON
artechhouse.com

Library of Congress Cataloging-in-Publication Data
A catalog record for this book is available from the U.S. Library of Congress.

Cover design by Joi Garron

ISBN-13: 978-1-63081-987-3

Accompanying appendices can be found at: https://quaxys.com/book.

685 Canton Street
Norwood, MA 02062

10 9 8 7 6 5 4 3 2 1

To my beloved parents, whose unwavering guidance and wisdom have illuminated my path and shaped me into the person I am today.

To the educators who have inspired, challenged, and supported me throughout my academic and professional journeys.

Contents

Foreword

Despite the almost infinite supply of information at our disposal today, we still need books. The author of a scientific book becomes a curator of knowledge, pulling together that which is important for their subject. Such curation is certainly not possible in the first moment after discovery; a process must take place in which knowledge, which has been gained often haphazardly and with great struggle, acquires some recognizable structure.

Quantum computers have been with us, intellectually speaking, for several decades now. But their physical realization has been slow and painful, and so the pedagogy of quantum computers has been, necessarily, also slow to emerge. It has taken decades for us to know what we need to know.

Alan Salari's book thus arrives, in a pioneering spirit, with an effort to define the intellectual landscape of the subject. It concerns itself with just one part of the vast subject, namely superconducting quantum computing. This is unquestionably a major branch of the subject, one which has a chance of being the dominant branch—maybe all the quantum computers of the future will be of this form. This is by no means certain, but it is best to be prepared for this outcome.

While some researchers 25 years ago understood that the superconducting device could be the realization of a qubit, they had no inkling of what would have to surround these qubits to make the quantum computer function. About 15 years of exploratory work followed. In the last 10 years, a consensus has gradually been appearing of how the superconducting quantum processor will really be built.

The big message from this consensus is that the superconducting quantum computer needs a lot of advanced techniques from cryogenic microwave engineering. This is the premise that drives Alan Salari's book, and is the featured item in the landscape that he depicts. We see here a cornucopia of microwave components, functions, and methodologies. Salari has covered in great detail all the microwave art that is seen as essential for the functioning of the quantum computer, and some art that is perhaps not yet fully in use—the state of system engineering is still in flux in this young field.

Why is such a large collection of high-performance, high-precision microwave techniques needed for the job? As an answer, I would like to offer an overall assessment of what must be done, by quoting from the opening pages of the book by theoretical physicist N. David Mermin, *Quantum Computer Science: An Introduction* (Cambridge University Press, 2007):

> It is tempting to say that a quantum computer is one whose operation is governed by the laws of quantum mechanics. But since the laws of

> quantum mechanics govern the behavior of all physical phenomena, this temptation must be resisted. Your laptop operates under the laws of quantum mechanics, but it is not a quantum computer. A quantum computer is one whose operation exploits certain very special transformations of its internal state, whose description is the primary subject of this book. The laws of quantum mechanics allow these peculiar transformations to take place under very carefully controlled conditions.
>
> In a quantum computer the physical systems that encode the individual logical bits must have *no physical interactions whatever that are not under the [almost] complete control of the program*.

I feel that this statement provides an excellent guide to the formidable job that the microwave engineer must perform in the constructing of the quantum computer. The insertion of the word "almost" is mine, but I think that it is in the spirit of the serious error analysis that Mermin provides later in his book, and as summarized also by Salari. Without "almost," there is no engineering job to be done, as complete control is a clearly impossible task and not worth attempting. But "almost complete control," the degree of which is made precise by the quantum theory, is what drives the unheard-of precision and performance requirements that the microwave system must achieve.

I hope that some of you readers, especially from the engineering disciplines, will be infected by the pioneering spirit, and use the knowledge gained to push the frontiers further toward a fully functioning quantum computer.

David P. DiVincenzo
January 2024

Preface

The first quantum revolution, which took place in the early twentieth century, transformed our understanding of the fundamental nature of matter and energy. This profound shift in our knowledge paved the way for developing groundbreaking technologies, such as transistors, atomic clocks, and lasers.

Today, we are witnessing the second quantum revolution, which seeks to push the boundaries of what we can achieve in controlling and manipulating quantum systems. The second quantum revolution aims to achieve unprecedented levels of precision and capabilities, leading to revolutionary applications in areas such as quantum computing, communication, sensing, and simulation. These remarkable advancements are poised to transform various fields, from telecommunications and cryptography to machine learning, finance, materials science, chemistry, and biology.

Quantum computing is one of the most exciting and challenging fields in the realm of quantum technologies. Among the various approaches to quantum computing, superconducting quantum computers have emerged as a promising technology. Superconducting quantum computing is a rapidly developing field that requires a multidisciplinary approach bringing together expertise in quantum physics, microwave engineering, cryogenic engineering, nanofabrication, material science, and software development.

The talent gap in quantum technologies has been challenging, requiring more educational resources, investments, and public awareness. While there has been much focus on training for developing quantum algorithms and software, there is a shortage of dedicated resources for understanding quantum hardware. This shortage motivated me to write this book to bridge the gap in the literature and to provide a resource for physicists and microwave engineers covering the theoretical principles and practical implementation of hardware for superconducting quantum processors.

This book covers a wide range of topics, from the basic principles of quantum mechanics, quantum computing, and superconducting qubits to the analysis of microwave systems and components, principles of cryogenics, and hardware implementation of superconducting quantum processors. From the outset, striking a balance between content that is accessible to both physicists and microwave engineers has been a significant challenge. In many cases, I have had to provide details that may seem obvious to one group but not the other. Despite these difficulties, I have strived to present the material in a clear and engaging way for all readers. Depending on the student's background and the instructor's preference, the material presented in this book can be used in both advanced undergraduate and graduate-level courses.

Chapters 1 and 2 introduce the principles of quantum mechanics and quantum computing, which are essential for understanding the operation of superconducting qubits. Chapter 3 delves into the details of superconducting qubits, discussing superconductivity, the principles of superconducting qubits, and control and readout techniques for single- and two-qubit gates. Chapters 4–6 cover the principles of microwave systems, microwave components, and electromagnetic compatibility, which are crucial for designing and optimizing the performance of superconducting quantum computers. Chapter 7 details cryogenic and room-temperature control hardware, collecting the knowledge from Chapters 1–6 to understand the hardware setup for the control and readout of superconducting qubits. Chapter 8 covers the principles of cryogenics, focusing on the dilution refrigeration technique.

A set of problems with a solutions manual, as well as a set of PowerPoint slides for instructors, is available through the following website: https://quaxys.com/book.

The second quantum revolution promises to unlock entirely new capabilities and drive transformative breakthroughs that were once thought impossible. I sincerely hope this book will provide a valuable resource for students, researchers, and professionals seeking to understand the intricacies of quantum hardware and, ultimately, help accelerate the development and adoption of this revolutionary technology.

Acknowledgments

I am deeply grateful for the unwavering support and invaluable assistance provided by numerous individuals throughout this book's writing, reviewing, and production. I would especially like to extend my heartfelt appreciation to David DiVincenzo for his overarching insight and thoughtful foreword.

Furthermore, I would like to acknowledge the contributions of the following individuals who meticulously reviewed my work:

Ankur Agrawal (Amazon Web Services);

Marc Almendros (Keysight Technologies);

Andy Cobin (Quantum Microwave);

David DiVincenzo [Forschungszentrum Jülich, Institute for Quantum Information (IQI) RWTH-Aachen];

Danilo Erricolo (University of Illinois, Chicago);

Yale Eckert (Rohde and Schwarz);

Giuseppe Esposito [Italian National Research Council (CNR); Institute for Electromagnetic Sensing of the Environment (IREA)];

Allen Fernandez (Jacobs);

Steven Frankel (Technion-Israel Institute of Technology);

Dean Friesen (Boeing);

Eric Holland (Keysight Technologies);

Alex Khan (ZebraKet);

Saesun Kim (NASA, Jet Propulsion Laboratory);

Piotr Kulczakowicz (Quantum Startup Foundry, University of Maryland, College Park);

James Liu (Intellian Technologies);

Dennis Lucarelli (Error Corp.);

Luke Mauritsen (Montana Instruments, member of the National Quantum Initiative Advisory Committee to the White House);

Brian McMenamin (Rohde and Schwarz);

Kevin Osborne (University of Maryland, College Park);

Matej Stefanac (Yazaki Group Europe);

Sara Sussman (Princeton University);

Oscar Viyuela (MIT, Harvard);

Matija Zesko (McKinsey);

Silvia Zorzetti (Fermi National Laboratory).

I am beyond grateful for their support and willingness to share their knowledge and insight.

Last, I am much obliged to Natalie McGregor, David Michelson, Julia Sayger, and Isabel Fowler, whose expert management and guidance were instrumental in the successful publication of this book.

CHAPTER 1

Introduction to Quantum Physics

> What we observe is not nature itself, but nature exposed to our method of questioning.
>
> —*Werner Heisenberg*

This chapter focuses on the principles of quantum mechanics essential for understanding the fundamental concepts of quantum computing and the operation of superconducting qubits. First, Section 1.1 offers a brief historical overview. Next, the chapter explores the Schrödinger equation and its applications in solving basic yet insightful quantum mechanical problems, such as the quantum harmonic oscillator, a two-level quantum system, and the quantum potential well. The chapter also delves into important quantum mechanics concepts including Heisenberg's uncertainty principle, entanglement, coherence, and quantum measurement.

1.1 A Brief History of Quantum Mechanics

The theory of quantum mechanics emerged in the 1920s to explain the behavior of the smallest particles in nature, such as electrons, atoms, and protons. At the quantum level, the laws of nature vastly differ from those that govern our everyday, macroscopic world. Quantum mechanics introduces concepts such as wave-particle duality, superposition, and entanglement, which have no classical counterparts. Despite these differences, classical mechanics and quantum mechanics are not entirely disconnected. According to the correspondence principle, quantum mechanics converges toward classical mechanics when the quantum numbers involved become large. For instance, in the case of the quantum harmonic oscillator, we can observe classical harmonic oscillator behavior for large quantum numbers, as discussed in Section 1.5.2.3.

The quantization of electromagnetic energy was crucial to the development of quantum mechanics. In the late nineteenth century, classical physics predicted the Rayleigh-Jeans catastrophe, which referred to an ideal black body in thermal equilibrium emitting an unbounded amount of energy as the wavelength approached the ultraviolet range. Planck resolved this issue by proposing that electromagnetic radiation could only be emitted or absorbed in discrete energy packets [1]. Each packet carries an energy E, which is given by $E = hf$, where h represents Planck's constant, and f represents the frequency of the electromagnetic wave. Planck's quantum of electromagnetic radiation was later named the photon by Einstein, who postulated that a photon was a physical particle. This assumption laid the groundwork for the wave-particle duality of electromagnetic radiation, whose wave nature had already been proved by Maxwell.

In 1913, Bohr made a significant breakthrough with his atomic model, which helped explain the Rydberg formula for the spectral emission lines of the hydrogen atom. To develop this model, Bohr proposed that the angular momentum L of the atom was quantized and could only take on values that were integer multiples of $\hbar = h/2\pi$, written as $L = n\hbar$ [2]. The discovery of the angular momentum's quantization was further reinforced by the Stern-Gerlach experiment of 1922, which demonstrated that electron spin was also quantized. This experiment provided additional evidence for the quantized nature of atomic particles [3].

In 1924, Louis de Broglie proposed the theory that matter has both particle and wave-like behavior, holding that particles with mass m and momentum p also behave as a wave with wavelength $\lambda = h/p$. Figure 1.1(a) shows the wave-like nature of an electron. These matter waves are a manifestation of the wave-particle duality in quantum mechanics. De Broglie connected the requirement that the angular momentum of an electron be an integer multiple of $\hbar$ to a standing-wave condition. In this interpretation, the electron is represented by a wave, and the number of wavelengths that fit along the circle of the electron's orbit must be an integer. The dual nature of matter was experimentally verified by Davisson and Germer, who showed that electrons scattered by the surface of a nickel's crystal displayed a diffraction pattern, a wave phenomenon. Later, in 1957, Claus Jönsson demonstrated the wave behavior of electrons using the double-slit experiment depicted in Figure 1.1(b).

In 1925, Werner Heisenberg, Max Born, and Pascual Jordan introduced matrix mechanics, which was the first formulation of quantum mechanics that was conceptually self-contained and mathematically consistent. In their theory, the physical properties of particles were represented as matrices that evolve over time. Around the same time, Erwin Schrödinger developed wave mechanics to describe the state of a quantum system by a wave function. Born interpreted the wave function as the probability density of finding a particle in a particular position at a particular time. Both matrix and wave mechanics were successful in reproducing the outcomes of previous theories, including Bohr's atomic model.

In 1928, Paul Dirac accomplished a major achievement by developing a theory combining quantum mechanics and special relativity. The Dirac equation is

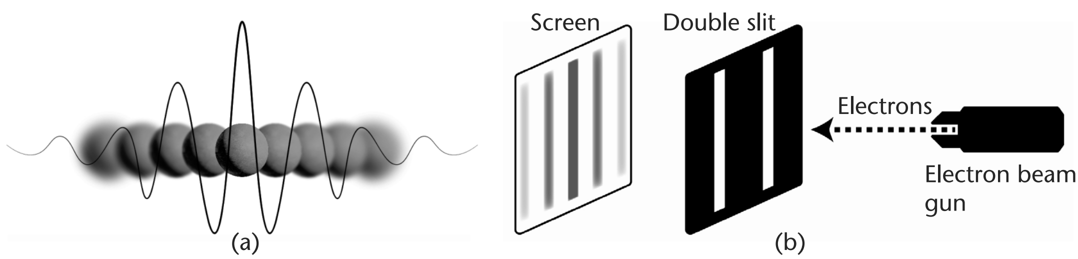


Figure 1.1 (a) The wave-like nature of electrons in quantum mechanics allows them to exist simultaneously at several locations. (b) Double slit experiment with electrons: An electron gun generates a stream of electrons, passing through the slits and interfering with one another to create an interference pattern on the screen.

a relativistic wave equation that describes the behavior of spin-1/2 particles, such as electrons and quarks, in their free form or under electromagnetic interactions. This equation was instrumental in explaining the electron gyromagnetic ratio and calculating the atom's spontaneous emission coefficients.

Through the end of the 1940s, advancements in microwave technology enabled more accurate measurements of the electron magnetic moment and the shift of the hydrogen atom's levels, known as the Lamb shift. These measurements revealed inconsistencies that Dirac's theory could not explain. To address these issues, Shinichiro Tomonaga, Julian Schwinger, and Richard Feynman developed quantum electrodynamics (QED), a relativistic quantum field theory of electrodynamics that describes how light and matter interact. QED, the first theory to fully reconcile special relativity with quantum mechanics, has made incredibly precise predictions, such as the electron's anomalous magnetic moment and the Lamb shift of hydrogen's energy levels. Feynman called QED "the jewel of physics" for its remarkable accuracy [4]. As discussed in subsequent chapters, the foundation of quantum computing is the interaction between light and matter, where laser light or microwave radiation interacts with atoms, ions, or quantum circuits to perform quantum computation.

A fundamental grasp of linear algebra and differential equations is essential to comprehending the material presented in this chapter.

1.2 Quantum Versus Classical Mechanics

The key distinction between classical and quantum mechanics lies in the inherently probabilistic nature of quantum mechanics. While statistical approaches are employed in classical mechanics due to incomplete knowledge of all degrees of freedom in a system, in quantum mechanics, the probabilistic nature is inherent and not just a reflection of our limited information about the system. In other words, the statistical nature of a quantum system persists even if we have precise information about all the system's degrees of freedom. This probabilistic nature is encoded in the wave function, which is the solution to the Schrödinger equation. The wave function enables us to calculate probabilistic quantities, such as expectation values, and variances of physically measurable quantities, such as position and momentum.

The principle of reversibility of physics laws is crucial to understanding the differences between the probabilistic characteristics of quantum and classical physics. This principle asserts that if a physical system is allowed to evolve from its initial state and then reversed precisely for the same amount of time, the system will return to its initial state. However, in classical systems where probability is involved, the system may not revert to its initial state. If we possess complete information about all the degrees of freedom, we can reverse every evolution of a classical system.

In quantum systems, the laws of physics are reversible, even in the presence of probability. Unlike in classical systems, this probability is inherent and does not result from a lack of information about the system. However, there is a crucial condition for the reversibility of the laws of quantum mechanics. Reversibility is possible only if we do not perform any measurement on the quantum system, allowing it to travel backward in time without any intervention. Following this condition, the quantum system will eventually return to its initial state. The measurement concept is another key distinction between classical and quantum physics, as discussed in Section 1.6. When a quantum system is measured, its state changes, leading to the loss of information through wave-function collapse. Therefore, we must avoid measuring the quantum system while checking for its reversibility.

Let us consider an example to better understand the fundamental concept of reversibility. Consider a system oscillating between two states, such as a binary pendulum with states L and R. If we start from state R, the system will return to R after two oscillations (RLRLR). However, suppose there is a small probability that the system randomly stays at state L. In this case, if the system stays at L during the first oscillation, we get RLLRL. When we run the system backward in time, assuming that the event of staying at L does not occur, we get LRLRL and end up in L state instead of returning to the initial state R. This example shows that the reversibility of physical laws does not apply to classical systems when probabilities are involved.

The conservation of information is crucial for the reversibility of physical processes. Chapter 2 covers quantum algorithms, which consist of a series of time-evolving quantum processes known as quantum gates. The reversibility of these gates is a critical requirement in quantum computing, ensuring that any operation on a normalized quantum state preserves the sum of probabilities of all possible outcomes to 1 to prevent the loss of quantum information. As discussed in Section 2.1.2, the reversibility in quantum mechanics is mathematically expressed through unitary transformations.

1.3 Schrödinger Equation

The laws of physics describe how the state of a system evolves over time. The state of a system can be characterized by various physical quantities, such as position, momentum, and energy. If the initial state of a system is known, we can use the appropriate laws to predict its future state. For instance, in classical mechanics, Newton's second law can be used to predict the position $x(t)$ of a particle at any time, given the initial conditions and the force acting on the particle. From the position $x(t)$, other quantities like velocity $v = dx/dt$, momentum $p = mv$, and kinetic energy $T = mv^2/2$ can be determined.

The Schrödinger equation provides a mathematical framework for quantum systems to calculate measurable quantities such as position, momentum, and energy. In the Schrödinger picture, the state of a quantum system is represented by the wave function $\Psi(x, t)$, which encodes the probabilities of all the possible outcomes. The wave function is used to calculate measurable physical quantities, such as position and momentum, which are known as observables. The Schrödinger equation is written as follows [5]:

$$\left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x,t)\right]\Psi(x,t) = i\hbar\frac{\partial}{\partial t}\Psi(x,t) \tag{1.1}$$

where m represents the mass of the particle, $V(x, t)$ is the potential, and $\hbar$ is the reduced Planck's constant. Alternatively, (1.1) can also be written in terms of Hamiltonian operator $H = -(\hbar^2/2m)(\partial^2/\partial x^2) + V(x, t)$ as $H\Psi(x, t) = i\hbar(\partial/\partial t)\Psi(x, t)$. Initially, the interpretation of the wave function $\Psi(x, t)$ was unclear. It was later proposed by Born that the square of the absolute value of the wave function, $|\Psi(x, t)|^2$, can be interpreted as the probability density of finding the particle at a particular location at a particular time.

The probability that a particle with wave function $\Psi(x, t)$ will be found between positions x and $x + dx$ at the time t is given by $|\Psi(x, t)|^2 dx$.

The wave function is more than just a probability function. Its phase is critical in enabling interference effects, a wave phenomenon, so referring to it as a wave function rather than a probability function is appropriate. Interference effects are essential for quantum algorithms to arrive at the correct answer, as illustrated in Section 2.2.5. It is important to note that the absolute phase of a wave function cannot be measured in experiments. However, the relative phase of two probability amplitudes can be determined by observing interference effects.

For a state to be physically realizable, the wave function must be square-integrable to ensure that the probability of finding a particle at any possible location adds up to one. This condition is known as normalization and is expressed mathematically as $\int_{-\infty}^{\infty} |\Psi(x,t)|^2 dx = 1$. In addition to being normalized, the wave function needs to fulfill the following mathematical properties [4]:

- The wave function must be continuous and single-valued everywhere.
- Derivative of the wave function is always continuous and single-valued except at a boundary where the potential $V(x, t)$ is infinite.
- The wave function must be normalizable. This requires $|\Psi(x, t)|^2$ to approach zero fast enough as $x \to \pm\infty$ so that $\int_{-\infty}^{\infty} |\Psi(x,t)|^2 dx$ remains finite.

1.4 The Machinery of Quantum Calculations

Before delving further into the mathematics of quantum mechanics, it is insightful to examine the structure of a quantum mechanical calculation. Compared to classical equations such as Maxwell's or Newton's equations, the way the Schrödinger equation is applied to calculations on observables, such as position and momentum, is different due to the probabilistic nature of quantum mechanics.

Figure 1.2 illustrates the process of quantum calculations. The first step is solving the Schrödinger equation to obtain the wave function. To accomplish this, we need to know the initial conditions, boundary conditions, and the potential function $V(x, t)$. It is worth noting that the Schrödinger equation can only be solved analytically for a limited number of cases. Therefore, numerical methods are commonly used to solve the Schrödinger equation for real-world problems. It is essential to mention that a wave function has a mathematical structure that falls under the concept of vector space.

In quantum mechanics, observables such as position and momentum are represented by operators. The operator and wave function is used to calculate the expectation value of the observable, indicating the most likely outcome when

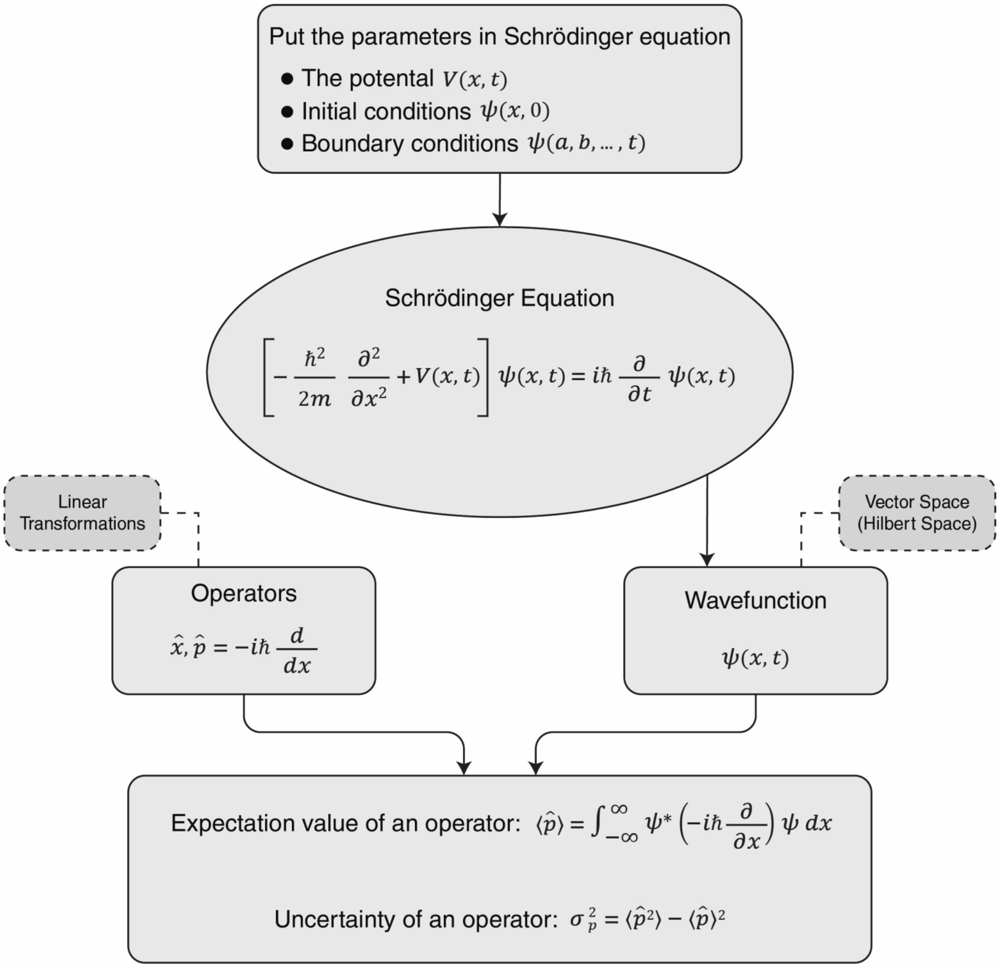


Figure 1.2 The machinery of quantum calculations.

we measure it. Due to the probabilistic nature of quantum mechanics, we work with expectation values instead of the deterministic outcomes known in classical mechanics. Additionally, we can calculate the uncertainty, which is the variance of the probability distribution. We will discuss these concepts in more depth later in Sections 1.6.2 and 1.6.3.

1.5 Solving the Schrödinger Equation

This section covers the time-independent Schrödinger equation and standard Hamiltonians, such as the potential well, potential barrier, and free particle. This section concludes with an examination of the quantum harmonic oscillator and two-level quantum systems, also known as spin 1/2 or qubits, essential for understanding the superconducting qubits.

1.5.1 Time-Independent Schrödinger Equation

If the potential function in the Schrödinger equation is time-independent, the wave function can be expressed as a product of a spatial function $\psi(x)$ and a temporal function $f(t)$ [i.e., $\Psi(x, t) = \psi(x)f(t)$]. This method is called the separation of variables. If we substitute the solution $\Psi(x, t)$ back in the Schrödinger equation (1.1), we obtain $\partial\Psi/\partial t = \psi(df/dt)$, $(\partial^2\Psi/\partial x^2) = (d^2\psi/dx^2)f$. Now the Schrödinger equation reads

$$i\hbar\frac{1}{f}\frac{df}{dt} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V \tag{1.2}$$

The left side of the (1.2) is only a function of t, and the right is only a function of x. This is only possible if both sides are constant, where we call this constant E, then $i\hbar(1/f)(df/dt) = E$, where E represents the energy of the system. The solution of this ordinary differential equation is straightforward $f(t) = Ce^{-iEt/\hbar}$

We can absorb the constant C into ψ and only use the $e^{-iEt/\hbar}$ term as the time solution. If we substitute $i\hbar(1/f)(df/dt) = E$ back in (1.2) and multiply both sides by ψ, we obtain the time-independent Schrödinger equation with Hamiltonian H

$$H\psi = E\psi \tag{1.3}$$

The $\psi(x)$ function can be obtained by solving the time-independent Schrödinger equation. The states given by solving (1.3) are called stationary states. The general solution can be written as a product of the stationary states $\psi(x)$ and the time evolution function as $\Psi(x, t) = \psi(x)e^{-iEt/\hbar}$.

Note that the probability density is time-independent and depends only on the stationary states $|\Psi(x, t)|^2 = \Psi^*\Psi = \psi^* e^{+iEt/\hbar}\psi e^{-iEt/\hbar} = |\psi(x)|^2$.

The general solution to the Schrödinger equation consists of a linear combination of stationary states, where the time evolution of each state is dependent on the energy of that state, as can be seen in the exponent of the temporal function $\Psi(x, t) = \sum_{n=1}^{\infty} c_n\psi_n(x)e^{-iE_nt/\hbar}$.

This profound result reveals that a quantum state can be expressed as a linear combination, or superposition, of multiple states. In other words, a quantum system can exist in multiple states simultaneously. The concept of superposition is one of the fundamental features of quantum mechanics, which makes quantum computing so powerful. The ability to exist in multiple states simultaneously allows quantum computers to perform computations in parallel, as explored in Chapter 2.

The relative phase of the wave functions plays a crucial role in interference effects, which is utilized in quantum algorithms to obtain the correct answer. To illustrate this, let us consider an example. As we will see later, the state of a quantum system can be represented using the Dirac notation, also known as bra-ket notation. Suppose that the state $|\psi_1\rangle$ is an equal superposition of the $|0\rangle$ and $|1\rangle$ qubit states $|\psi_1\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. And suppose that $|\psi_2\rangle$ is also an equal superposition of $|0\rangle$ and $|1\rangle$ states but that the two states are out-of-phase $|\psi_2\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$. When we examine the states $|\psi_1\rangle$ and $|\psi_2\rangle$ individually, the probabilities of both states being in $|0\rangle$ and $|1\rangle$ are equal. As a result, the relative phase between the superimposed states does not matter. However, if we create a new superposition $|\psi_3\rangle = (|\psi_1\rangle + |\psi_2\rangle)/\sqrt{2}$, where the system has an equal probability of being in either $|\psi_1\rangle$ or $|\psi_2\rangle$ state, the relative phase between the two superposed states becomes crucial. This is because the phases of the two states can interfere constructively or destructively, leading to different outcomes when the system is measured. We can see that by substituting equations $|\psi_1\rangle$ and $|\psi_2\rangle$ into superposition $|\psi_3\rangle = (|\psi_1\rangle + |\psi_2\rangle)/\sqrt{2}$, we get $|\psi_3\rangle = |0\rangle$. This means that the $|1\rangle$ state vanished due to the destructive interference between the two states.

Section 1.5.2 examines some standard Hamiltonians to gain more insight into the solutions of the Schrödinger equation.

1.5.2 Standard Hamiltonians

While the Schrödinger equation can only be solved analytically for a few cases, these solutions offer valuable insights. This section explores the solutions for the free particle, potential well, potential barrier, harmonic oscillator, and two-level quantum systems.

1.5.2.1 Free Particle

For a free particle, the potential $V(x, t) = 0$. The Schrödinger equation reads $-(\hbar^2/2m)(d^2\psi/dx^2) = E\psi$ or $d^2\psi/dx^2 = -k^2\psi$, where $k \equiv \sqrt{2mE}/\hbar$. The general solution of this equation consists of two traveling waves $\psi(x) = Ae^{ikx} + Be^{-ikx}$. The first term in $\psi(x) = Ae^{ikx} + Be^{-ikx}$ represents a wave traveling to the right, and the second represents a wave traveling to the left. There is no constraint on the values of k since no boundary conditions are associated with the free particle. The energy of a free particle can take any value, and its energy spectrum is continuous. A superposition of plane waves describes the time-dependent solution $\Psi(x, t) = \psi(x)e^{-iEt/\hbar} = Ae^{ik(x-(\hbar k/2m)t)} + Be^{-ik(x+(\hbar k/2m)t)}$. The wave function of a free particle cannot be normalized,

$$\int_{-\infty}^{+\infty} \Psi_k^* \Psi_k \, dx = |A|^2 \int_{-\infty}^{+\infty} 1 \, dx = |A|^2 (\infty) \tag{1.4}$$

This proves that there is no such thing as a free particle with definite energy. However, we can obtain a meaningful physical interpretation by using a linear combination of solutions, as described by the superposition principle, where the resulting wave function can be normalized

$$\Psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \phi(k) e^{i\left(kx - \frac{\hbar k^2}{2m}t\right)} dk \tag{1.5}$$

Equation (1.5) bears a resemblance to the Fourier transform, where the coefficients $\phi(k)$ can be computed using the inverse Fourier transform

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \Psi(x,0) e^{-ikx}\, dx \tag{1.6}$$

Equation (1.6) represents a range of energies that correspond to each value of k. This results in a localized and coherent superposition of plane waves, which is known as a wave packet [see Figure 1.3(a)]. The size and shape of the wave packet change as it propagates [Figure 1.3(b)].

As shown in Figure 1.3(a), two velocities are associated with a wave packet, the group velocity v_g and the phase velocity v_p. The phase velocity is related to the velocity of the ripples given by $v_{\text{phase}} = \omega/k$. The group velocity is related to the velocity of the envelope. The group velocity can be approximated as follows if the k has a small spread[1] $v_{\text{group}} = d\omega/dk$. The dispersion relation ($\omega - k$ relation) for a free particle is given by $\omega = (\hbar k^2/2m)$. The relations for the dispersion, group, and phase velocity lead to $v_{\text{classical}} = v_{\text{group}} = 2v_{\text{phase}}$.

1.5.2.2 Potential Well

Let us dive into the fascinating topic of potential wells. In classical mechanics, potentials can be engineered to manipulate the movement of particles, creating bound states or controlling scattering states. The same holds for quantum particles, but the possibilities are even broader in the quantum realm. For instance, unlike classical particles, a quantum particle can tunnel through a finite potential barrier, meaning that it can escape even if it lacks the required energy. This tunneling phenomenon allows for exciting applications in nanoelectronics, quantum computing, and even nuclear fusion.

An infinite square well with the following potential function is shown in Figure 1.4(a)

$$V(x) = \begin{cases} 0, & 0 \le x \le L \\ \infty, & \text{otherwise} \end{cases} \tag{1.7}$$

This potential function results in the following wave function after solving the Schrödinger equation [6] $\psi_n(x) = \sqrt{2/L}\,\sin(n\pi x/L)$ with energy levels $E_n = n^2\pi^2\hbar^2/2mL^2$.

In an infinite square potential well, wave functions exhibit a simple sinusoidal pattern: standing waves. These states are known as bound states since each state's energy is lower than that of the potential well, which, in this case, is infinite. This property makes potential wells ideal for quantum mechanical systems, as they can create stable and controllable environments for experiments in fields like quantum information processing and quantum simulations. One of the most exciting applications of potential wells is particle trapping. For example, single electrons

1. For a large range of k's, the wave packet's shape changes rapidly since different wave components travel at a different speed; therefore, we're not dealing anymore with a well-defined group of waves with a well-defined velocity.

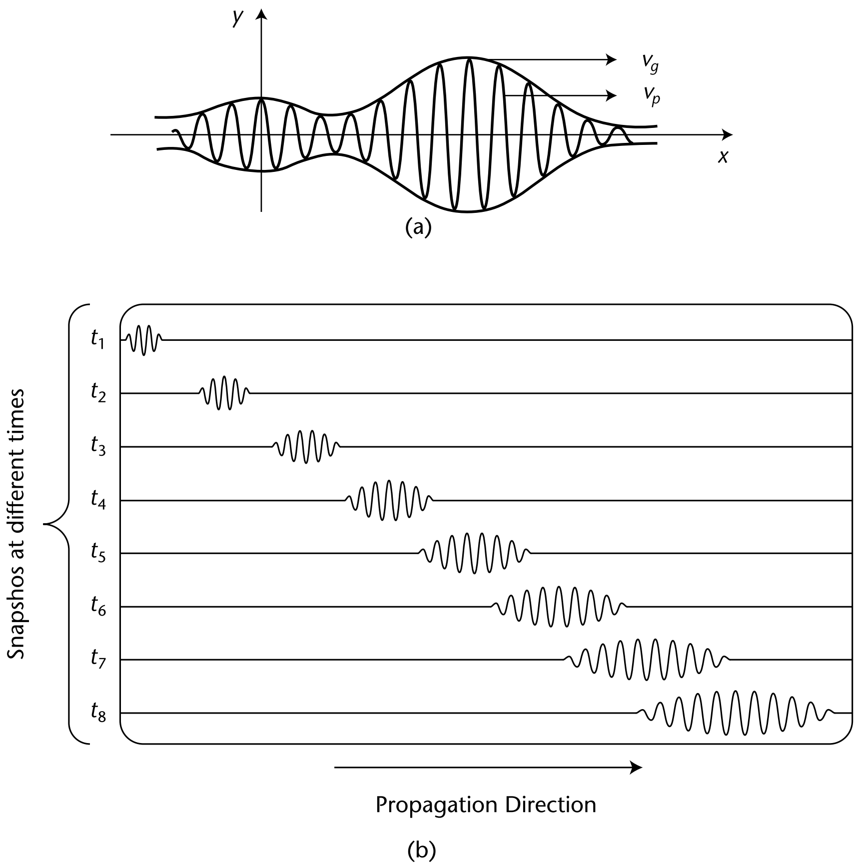


Figure 1.3 (a) A wave packet, where the envelope travels with a group velocity, whereas the ripples travel with the phase velocity, and (b) the spread of a wave packet; the shape and size of the wave packet change as it propagates.

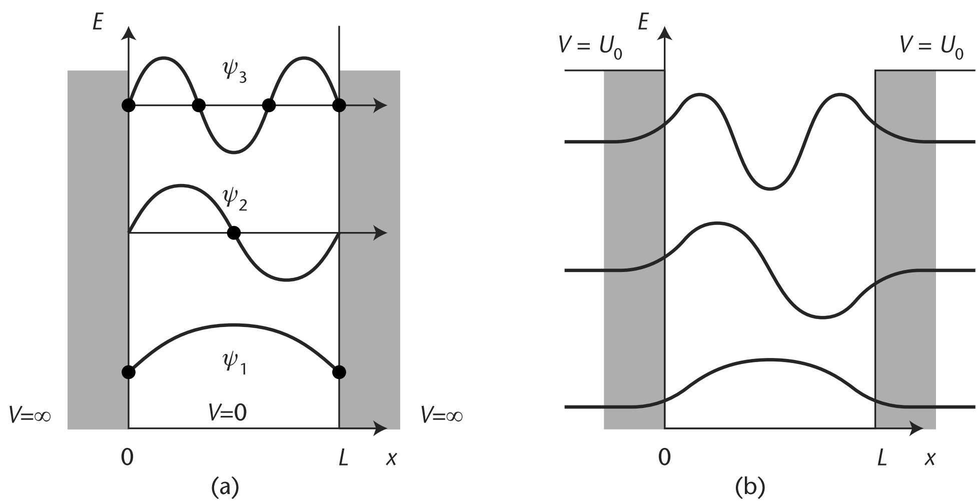


Figure 1.4 (a) An infinite square well with a width of *L* and (b) a finite square well, where the wave functions decay exponentially outside the well.

can be trapped in a spin qubit, or neutral atoms can be trapped using an optical potential well.

An interesting phenomenon called tunneling can occur in a finite square well, as shown in Figure 1.4(b). This is a characteristic of quantum mechanics, and it means that even when a particle's energy is lower than the surrounding potential, there is still a nonzero probability of finding the particle outside the well.

Quantum tunneling has many applications in semiconductor devices, including tunneling diodes and imaging techniques, such as the scanning tunneling microscope (STM). In addition, the tunneling effect is used for the spin-charge conversion readout of spin qubits. Josephson junctions, critical components in superconducting qubits, also exhibit quantum tunneling, as discussed in Chapter 3.

1.5.2.3 Harmonic Oscillator

The quantum harmonic oscillator is a fundamental concept in understanding and modeling quantum mechanical systems. It is used to model a wide range of physical systems, including electromagnetic waves, resonators, qubits, and molecular potentials near their minimum. In classical mechanics, a harmonic oscillator is a mass m attached to a spring with a force constant k. The motion of the system is governed by Hooke's law $F = -kx = md^2x/dt^2$, where the potential energy is given by $V(x) = kx^2/2$.

The equivalent problem in quantum mechanics is to solve the Schrödinger equation for the following potential $V(x) = m\omega^2x^2/2$, where ω is the angular frequency of the oscillator. To determine the system's energy levels, it is sufficient to solve the time-independent Schrödinger equation.

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}m\omega^2x^2\right)\psi = E\psi \tag{1.8}$$

We do not delve into the mathematical details of the solution, as it is readily available in practically all introductory books on quantum mechanics [5]. However, it is crucial to understand each state's energy levels and the wave functions of the harmonic oscillator. The solution to the Schrödinger equation is given by

$$\Psi_n(x) = \sqrt{\frac{1}{2^n n!}} \cdot \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \cdot e^{-\frac{m\omega x^2}{2\hbar}} \cdot H_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right) \tag{1.9}$$

$H_n(\sqrt{m\omega/\hbar}x)$ are Hermite polynomials. The Hermite polynomials and, consequently, the wave function is an odd function for odd values of n and an even function for even values of n, as shown in Figure 1.5(a). The energy of each state is given by

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \quad \text{for } n = 0,1,2,\ldots \tag{1.10}$$

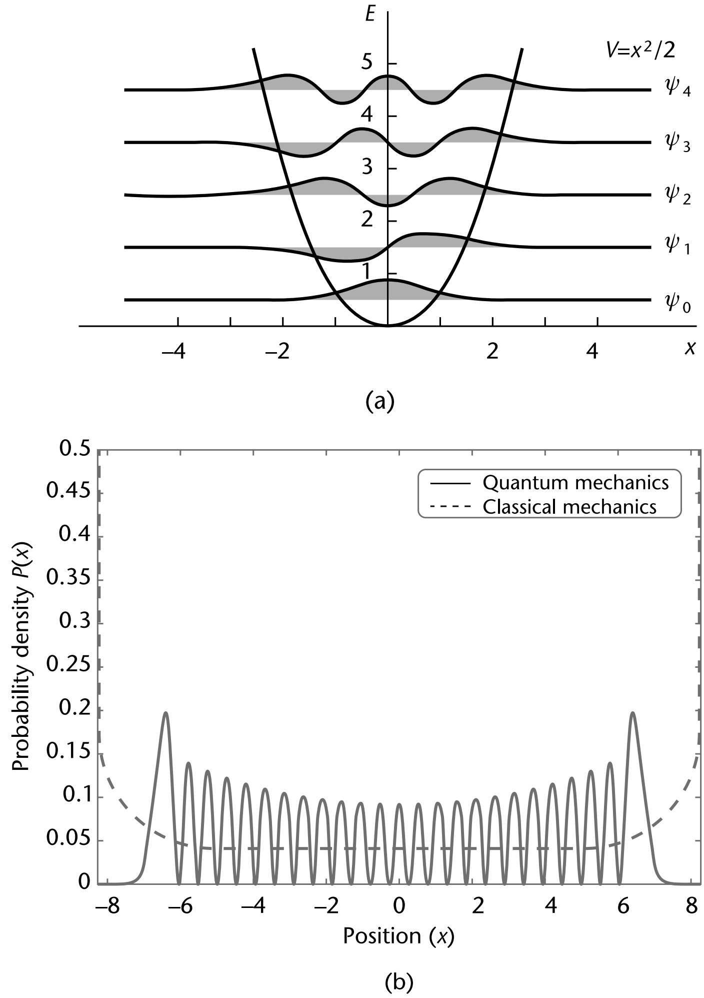


Figure 1.5 (a) Wave functions of a quantum harmonic oscillator. (b) When comparing a quantum harmonic oscillator to a classical one, we observe that the behavior of the quantum system becomes increasingly like that of a classical harmonic oscillator as the quantum numbers become large. The wave function of a quantum harmonic oscillator for n = 100 indicates this convergence.

There are several notable differences between quantum and classical harmonic oscillators, including the following.

- The lowest allowed harmonic oscillator energy $E_0 = \hbar\omega/2$ is not zero.
- The probability of finding the particle outside the classically allowed range is not zero [see Figure 1.5(a)].

Moreover, it is worth noting the following.

- The ground state of a harmonic oscillator is a minimum uncertainty state (see Section 1.6.3 for the concept of uncertainty), resulting in a Gaussian position-space wave function.

- All energy eigenstates of the harmonic oscillator obey the virial theorem. This theorem relates the kinetic energy T and potential energy V of the system through their expectation values, [i.e., $\langle T\rangle$ and $\langle V\rangle$, respectively]. (See Section 1.6.2 for the concept of expectation value.) $\langle T\rangle = \langle V\rangle = E_n/2$.

Despite all the differences between the classical and quantum harmonic oscillators, they behave similarly for large quantum numbers, as illustrated in Figure 1.5(b). This is an example of the correspondence principle introduced in Section 1.1.

1.5.2.4 Ladder Operators

Sometimes, instead of being interested in the wave function corresponding to each state, we seek to understand the transitions between these states. For instance, in a harmonic oscillator, we can assign a number to each state, such as $|0\rangle$, $|1\rangle$, $|2\rangle$, …, making a ladder of states [7]. Then, we can define ladder operators, which consist of raising and lowering operators acting on a state, moving it up or down the ladder of states. Before delving into this concept, let's have a quick look at the Dirac notation. The bra-ket or Dirac notation is widely used in quantum mechanics. The bra vector $\langle A|$ and the ket vector $|A\rangle$ are defined as $\langle A| = (A_1^* \quad A_2^* \quad \dots \quad A_N^*)$ and $|A\rangle = (A_1 \quad A_2 \quad \dots \quad A_N)^T$, respectively, where A_N^* is the complex conjugate of A_N. In quantum mechanics, the dagger symbol (†) denotes the adjoint or Hermitian conjugate of an operator or a state vector. The Hermitian conjugate, which is the transposed conjugate of a bra, is the corresponding ket, and vice versa (i.e., $|A\rangle^\dagger = \langle A|$ and $\langle A|^\dagger = |A\rangle$). Moreover, $|\phi\rangle = A|\psi\rangle$ if and only if $\langle\phi| = \langle\psi|A^\dagger$.

Now, let us explore how the ladder operators are constructed. Recall that the Hamiltonian of a harmonic oscillator is given by

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 \tag{1.11}$$

With a clever variable change, the Hamiltonian can be written in a more straightforward and useful form. We define $\hat{a}$ and $\hat{a}^\dagger$ operators as follows:

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right) \tag{1.12}$$

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right) \tag{1.13}$$

where a and $a^\dagger$ operators are called lowering and raising operators, respectively. The Hamiltonian in (1.11) can be expressed in terms of raising and lowering operators $H = (\hbar\omega/2)(a + a^\dagger)$. The following relations show how these operators act on the state $|n\rangle$

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle,\ \hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \tag{1.14}$$

So when $\hat{a}$ acts on the state $|n\rangle$, it brings it down by one level to the state $|n-1\rangle$. For the raising operator, it brings the state $|n\rangle$ one level up to $|n+1\rangle$. Section 1.5.2.4 use the concept of raising and lowering operators to show the transition between the ground and excited states of the qubit. The commutation relation between the raising and lowering operators is $[a, a^\dagger] = 1$.

The matrix form of the raising and lowering operators can be obtained by using $a^\dagger_{ij} = \langle \psi_i | a^\dagger | \psi_j \rangle$ and $a_{ij} = \langle \psi_i | a | \psi_j \rangle$. The eigenvectors ψ_i are those of the harmonic oscillator expressed in the number basis. The matrix form of the eigenstates of the harmonic oscillator $|n\rangle$ is given by $|0\rangle = [1 \ \ 0 \ \ \dots \ \ 0]$, $|1\rangle = [0 \ \ 1 \ \ \dots \ \ 0]$, and so on.

1.5.2.5 Quantum LC Oscillator

The concept of a quantum harmonic oscillator can be used to model a quantum LC oscillator, as depicted in Figure 1.6(a) [8, 9]. A qubit is an example of a quantum anharmonic oscillator due to the nonlinear term in its potential, as discussed in Chapter 3. Despite this, the harmonic oscillator can still accurately approximate the anharmonic oscillator at low energy levels, as demonstrated in Figure 1.6(b).

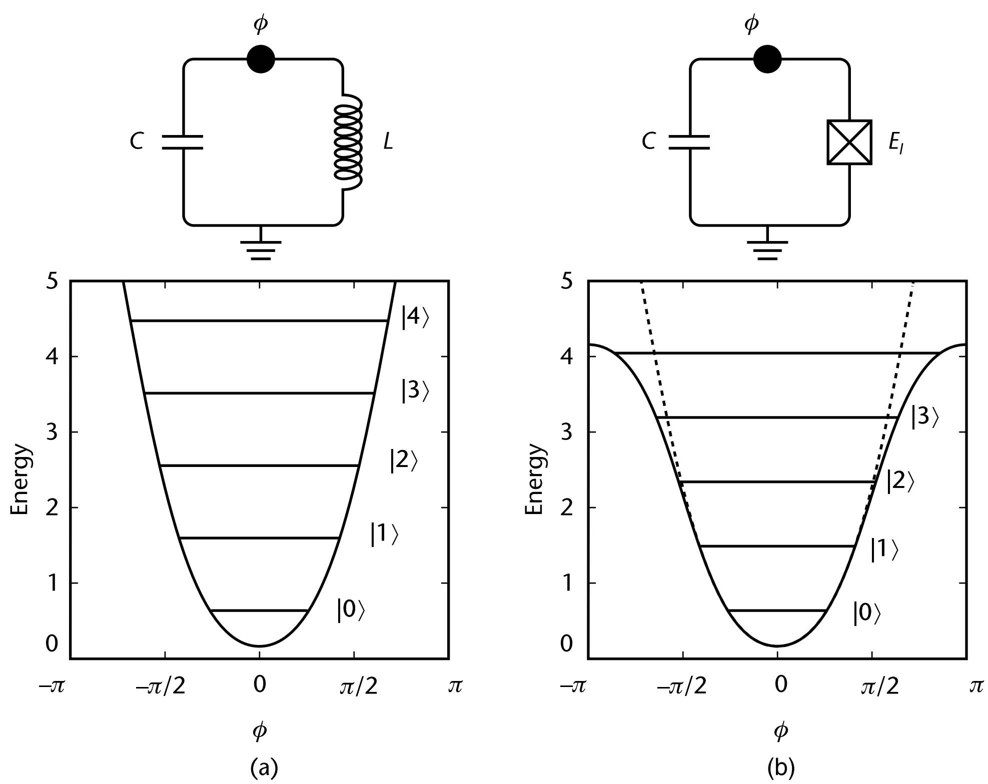


Figure 1.6 (a) An LC circuit can be modeled as a quantum harmonic oscillator. (b) A qubit consisting of a nonlinear element called a Josephson junction parallel to a capacitor is an anharmonic oscillator. However, a qubit can be approximated as a harmonic oscillator at low energy levels.

The Hamiltonian of an LC oscillator has a similar form to the quantum harmonic oscillator introduced

$$H = \frac{1}{2C}Q^2 + \frac{\varphi^2}{2L} \tag{1.15}$$

where C and L are the capacitance and inductance of the circuit, Q is the charge on the capacitor plates, and φ is the flux through the inductor. Table 1.1 compares the parameters of quantum electrical and mechanical oscillators.

The coordinate Φ and its conjugate momentum Q can be promoted to quantum operators obeying the canonical commutation relation $[\hat{Q},\hat{\Phi}] = -i\hbar$ [8, 9]. We can rewrite the Hamiltonian in terms of lowering and raising operators.

$$H = \frac{\hbar\Omega}{2}\left(a^\dagger a + aa^\dagger\right) = \hbar\Omega\left(a^\dagger a + \frac{1}{2}\right) \tag{1.16}$$

where $\Omega = 1/\sqrt{LC}$ is the resonant frequency and the raising $a^\dagger$, and lowering a operators can be expressed as

$$a = +i\frac{1}{\sqrt{2L\hbar\Omega}}\hat{\Phi} + \frac{1}{\sqrt{2C\hbar\Omega}}\hat{Q},\ a^\dagger = -i\frac{1}{\sqrt{2L\hbar\Omega}}\hat{\Phi} + \frac{1}{\sqrt{2C\hbar\Omega}}\hat{Q} \tag{1.17}$$

which obeys $[a, a^\dagger] = 1$. Chapter 3 discusses the two coupled LC oscillators to model the qubit-cavity coupling.

Table 1.1 Comparison of Quantum Electrical and Mechanical Oscillators

Electrical Oscillator	*Mechanical Oscillator*
Φ	x
Q	p
C	m
L^{-1}	k
$V = \dot{\Phi} = \frac{Q}{C}$	$v = \dot{x} = \frac{p}{m}$
$I = \dot{Q} = -\frac{1}{L}\Phi$	$F = \dot{p} = -kx$
$\omega_0 = \frac{1}{\sqrt{LC}}$	$\omega_0 = \sqrt{\frac{k}{m}}$
$H = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$	$H = \frac{p^2}{2m} + \frac{kx^2}{2}$
$\left[\hat{\Phi},\hat{Q}\right] = i\hbar$	$\left[\hat{x},\hat{p}\right] = i\hbar$

1.5.2.6 Two-Level Quantum System: Qubit

Assume an isolated two-level quantum system, also called a qubit, with the ground state $|g\rangle \equiv [1\ \ 0]^T$, the excited state $|e\rangle \equiv [0\ \ 1]^T$, and energy difference of $E = E_e - E_g = \hbar\omega_{eg}$. As the ground and excited states are orthogonal, we have $\langle e|g\rangle = 0$ and $\langle g|g\rangle = \langle e|e\rangle = 1$. The Hamiltonian of the qubit $\hat{H}_q$ is given by

$$\hat{H}_q = \frac{\hbar\omega_{eg}}{2}\left(|e\rangle\langle e| - |g\rangle\langle g|\right) = \frac{\hbar\omega_{eg}}{2}\sigma_z \tag{1.18}$$

where $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is the Pauli-Z matrix.

When this Hamiltonian acts on the ground state $|g\rangle$, it returns $\hat{H}_q|g\rangle = (\hbar\omega_{eg}/2)(|e\rangle\langle e|g\rangle - |g\rangle\langle g|g\rangle) = -(\hbar\omega_{eg}/2)|g\rangle$, and when it acts on the excited state $|e\rangle$, it returns $\hat{H}_q|e\rangle = (\hbar\omega_{eg}/2)|e\rangle$.

Like a quantum harmonic oscillator, lowering and raising operators can be defined for a qubit. The raising operator $\sigma^+ = |e\rangle\langle g|$ acting on the ground state brings the qubit to the excited state, and the lowering operator $\sigma^- = |g\rangle\langle e|$ acting on the excited state brings the qubit to the ground state. For a qubit, it is more common to represent the ground and excited states by $|0\rangle$ and $|1\rangle$, respectively.

1.5.2.7 Bloch Sphere

A two-level system's pure state $|\psi\rangle$ can be expressed as a superposition of basis vectors $|0\rangle \equiv [1\ \ 0]^T$ and $|1\rangle \equiv [0\ \ 1]^T$ (i.e., $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$), where the coefficients α and β are complex numbers and normalized $|\alpha|^2 + |\beta|^2 = 1$.

The Bloch sphere shown in Figure 1.7(a) is a helpful tool to visualize the state space of a qubit and has no equivalent for more than one qubit. Figure 1.7 compares the state space of a classical bit and a qubit. While a bit has only two possible states, a qubit can have any arbitrary state on the Bloch sphere.

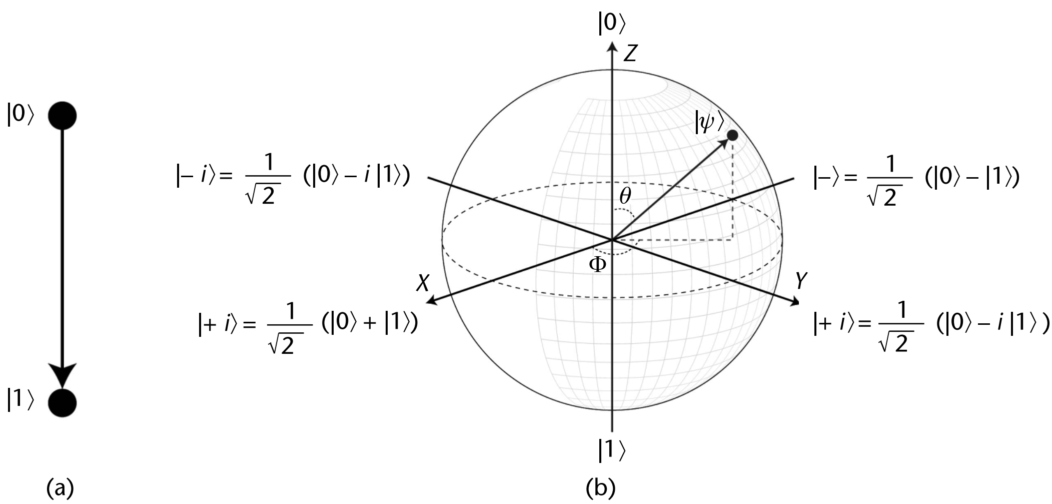


Figure 1.7 (a) State space of a bit and (b) state space of a qubit on a Bloch sphere.

We can associate a vector with the state of a qubit on a Bloch sphere. This vector extends from the origin to the surface of the Bloch sphere, which has a unit radius. Its orientation is determined by the ϕ and θ angles shown in Figure 1.7(a). Therefore, an arbitrary state $|\psi\rangle$ on a Bloch sphere can be expressed as follows [5]:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle \text{ where } 0 \leq \theta \leq \pi \text{ and } 0 \leq \phi \leq 2\pi \tag{1.19}$$

The relative phase between the basis states is encoded in the ϕ angle. As shown in Figure 1.7(a), the intersection of the Bloch sphere with the x and y axes represents various superposition states, while the intersection with the z-axis represents the $|0\rangle$ and $|1\rangle$ states.

1.6 Quantum Measurement

Early in the development of quantum mechanics, physicists and philosophers struggled with misunderstandings stemming from the theory's inherent probabilistic character, measurement process, and interpretation of measured data. In discussing the many interpretations of quantum mechanics and their consequences beyond the scope of this book, we cover one widely accepted and experimentally confirmed interpretation: the Copenhagen interpretation, associated with Bohr and proponents of this interpretation. According to this interpretation, the act of measurement results in the collapse of the wave function. Through measurement, we force the system to choose one state among all possible states and remain in that state afterward. The collapse of the wave function implies that passive observation is impossible, as the act of observing may alter what is being observed.

1.6.1 Collapse of the Wave Function

The concept of quantum measurement and the collapse of the wave function is depicted in Figure 1.8. The amplitude of the $|\Psi|^2$ function at each point represents the probability of finding the particle at that point. As shown in Figure 1.8, when a measurement is taken, the wave function collapses, and the particle is found at one single position and remains in that position afterward. Therefore, if we measure the particle after the collapse of the wave function, we will find the particle at the same position, regardless of how many times we repeat the measurement.

The question is how to perform a meaningful measurement of position or any other observable if the measurement results might change every time owing to the system's probabilistic character. The solution is to determine the expectation value of an observable. For instance, when measuring the position, we can prepare n identical quantum systems in the same state Ψ, as shown in Figure 1.8, and measure their positions. By taking an average of this position data, we can determine the position where we expect to find the particle the most. Naturally, the more measurement points we have, the closer we can get to the expectation value. Section 1.6.2 discusses how to calculate the expectation value mathematically.

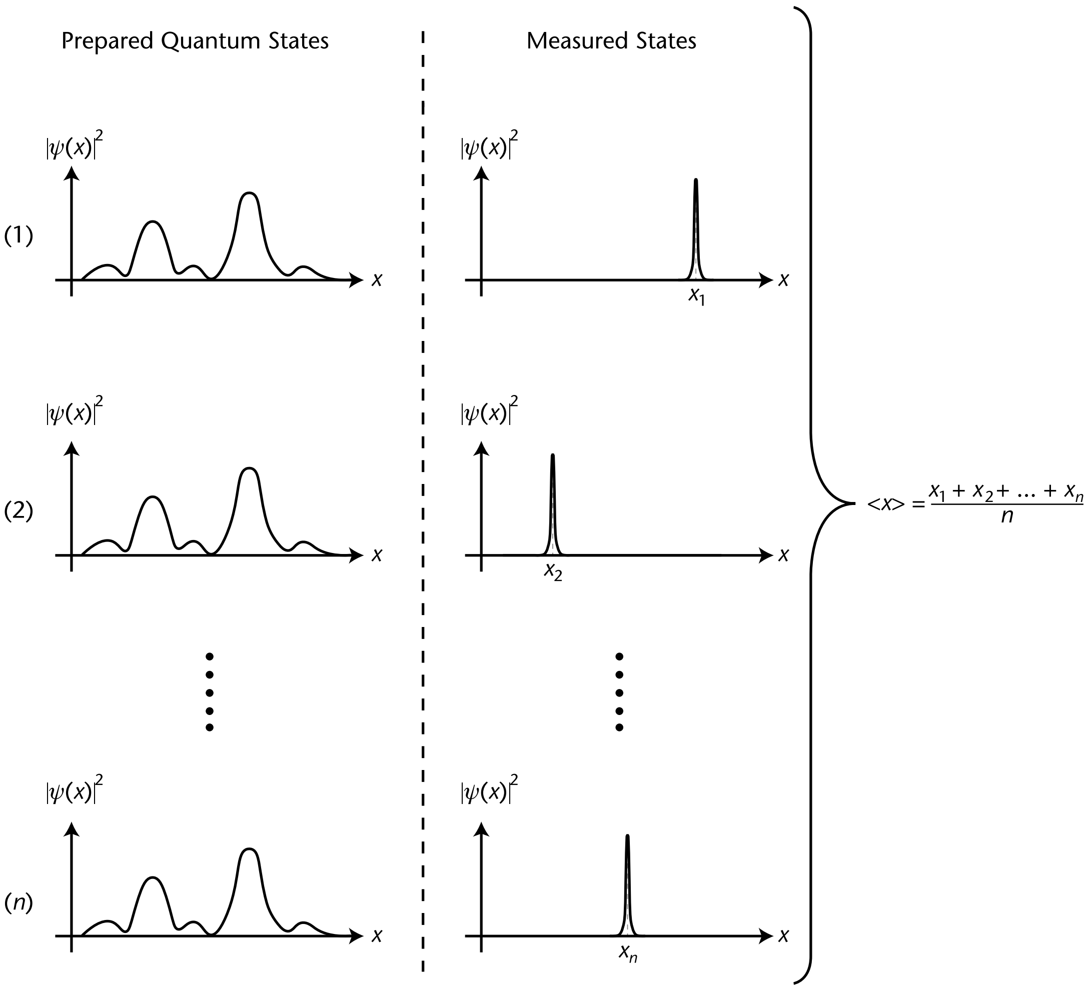


Figure 1.8 Left plots display the wave function, while the right plots demonstrate the wave function's collapse following measurement. This illustration presents the concept of expectation value in quantum measurements. We prepare 'n' identical quantum systems in the same quantum state and conduct measurements on each system. As a result of the probabilistic nature of quantum mechanics, every measurement yields a different outcome. The expectation value is defined as the average of these measured values. Consequently, we anticipate the particle to predominantly occupy the position $\langle x \rangle$.

The measured position values depicted in Figure 1.8 exhibit a certain distribution. This distribution can be created by plotting the measured positions on the x-axis and the frequency of each position on the y-axis. The standard deviation measures how much the values in a distribution deviate or spread out from the mean value, and the variance is referred to as the uncertainty of the position operator. Sections 1.6.2 and 1.6.3 cover the mathematical calculation of the expectation and uncertainty values.

1.6.2 Expectation Value

Recall from Section 1.4 that an operator in quantum mechanics represents an observable. The expectation value of an observable A with the operator $\hat{A}$ is given by

$$\langle A \rangle = \int_{-\infty}^{\infty} \psi^*(x,t)\hat{A}\psi(x,t)\,dx \tag{1.20}$$

If $\Phi(p, t)$ represents the momentum-space wave function, we have, in general,

$$\langle Q(x,p,t) \rangle = \begin{cases} \int \Psi^* \hat{Q}\left(x, \frac{\hbar}{i}\frac{\partial}{\partial x}, t\right)\Psi\,dx, & \text{in position space;} \\ \int \Phi^* \hat{Q}\left(-\frac{\hbar}{i}\frac{\partial}{\partial p}, p, t\right)\Phi\,dp, & \text{in momentum space.} \end{cases} \tag{1.21}$$

As shown in Section 1.5.2.2, the wave functions are $\psi_n(x) = \sqrt{2/L}\sin(n\pi x/L)$. We can use (1.22) to calculate the expectation value of a particle's position in an infinite square well in the ground state ($n = 1$).

$$\langle x \rangle = \int_0^L \sqrt{\frac{2}{L}}\sin\left(\frac{\pi x}{L}\right)x\sqrt{\frac{2}{L}}\sin\left(\frac{\pi x}{L}\right)dx = \frac{L}{2} \tag{1.22}$$

where the $u = \pi x/L$ change of variable is used to solve the above integral. It shows that we usually expect to find the particle in the center of the well for the ground state.

Let us calculate the momentum expectation value for a particle in its ground state in an infinite square well.

$$\langle p \rangle = \int_0^L \left(\sqrt{\frac{2}{L}}\sin\left(\frac{\pi x}{L}\right)\right)\left(-i\hbar\frac{d}{dx}\left(\sqrt{\frac{2}{L}}\sin\left(\frac{\pi x}{L}\right)\right)\right)dx = -ih\frac{L}{\pi}(0) = 0 \tag{1.23}$$

The expectation value of the momentum of a particle in an infinite square well is zero due to the standing waves that fit in the well. Standing waves oscillate in a place like a guitar string, meaning there is no net energy flow or momentum in either direction.

To investigate whether there is a connection between expectation values in quantum mechanics and classical mechanics, we can turn to Ehrenfest's theorem. This theorem states that, similar to Newton's second law, the time rate of change of momentum (i.e., the force) is equal to the gradient of the potential $d\langle p\rangle/dt = \langle -\partial V/\partial x\rangle$.

1.6.3 Variance or Uncertainty

The probabilistic nature of quantum systems results in measured values having a certain distribution. The standard deviation is used to express the spread of this distribution. The standard deviation of the position operator is defined as $\sigma_x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$ [5]. The variance or σ_x^2 is referred to as the uncertainty of an operator. Note that every calculation of the standard deviation σ_x or expectation value is done for a specific eigenstate.

Let us calculate the standard deviation for the position of a particle in the ground state of an infinite square well

$$\langle x^2 \rangle = \int_0^L \left(\sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)\right) x^2 \left(\sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)\right) dx \tag{1.24}$$

By using the change of variable $u = \pi x/L$ we obtain $\langle x^2 \rangle = 0.283L^2$, and therefore $\sigma_x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{0.283L^2 - (0.5L)^2} = 0.182L$.

The standard deviation of the momentum of a particle in the ground state of an infinite square well is calculated as follows $\sigma_p = \sqrt{(h^2/4L^2) - (0)^2} = h/2L$.

Section 1.6.4 explores the relationship between the uncertainties in position and momentum, as described by Heisenberg's uncertainty principle.

1.6.4 Uncertainty Principle

Recall from Section 1.1 that wave-particle duality is a significant property of quantum mechanics. Both wave and particle pictures can represent quantum physics processes equally well. However, there is a limit to the application of the particle picture to quantum systems dictated by the uncertainty principle. This implies that as one concept, such as the position, approaches the particle picture, the other, such as the velocity, becomes fuzzy and distant from its initial classical meaning. Let us consider an example to clarify this.

As explained in Section 1.5.2.1, a wave packet can be used to visualize the position of a quantum particle spread over the distance Δx, implying that the particle can be anywhere in the Δx range. The electron's velocity is associated with the group velocity of the wave packet, as mentioned earlier. Due to dispersion, the size and shape of the wave packet change as it moves; therefore, the group velocity cannot be precisely defined. As a result, the velocity also spreads over a range defined by Δv, which implies that the particle's velocity has a value that belongs to the Δv range. It is important to note that this uncertainty should be viewed as a fundamental aspect of the electron rather than a sign that the wave picture is not applicable.

The wave packet can be written as a superposition of plane sinusoidal waves with wavelengths centered around λ_0 with a spread of $\Delta\lambda$ [6]. The number of crests that fall within the Δx boundary is approximately $n = \Delta x/\lambda_0$. Outside the Δx boundary, the superposition of the waves must have a canceling effect. This is possible if, and only if, some of the component waves contain at least $n + 1$ waves that fall in the critical range. This phenomenon results in an interference pattern, where the waves constructively add up within the Δx boundary and destructively outside it. This is expressed mathematically as [10]

$$\frac{\Delta x}{\lambda_0 - \Delta\lambda} \geq n + 1 \tag{1.25}$$

If we substitute $n = \Delta x/\lambda_0$ in (1.25), then we obtain $(\Delta x/\lambda_0 - \Delta\lambda) \geq (\Delta x/\lambda_0) + 1$. After doing some algebra, we obtain $\Delta x \Delta\lambda/\lambda_0(\lambda_0 - \Delta\lambda) \geq 1$. Since $\lambda_0 \gg \Delta\lambda$, the following approximation can be applied $\lambda_0(\lambda_0 - \Delta\lambda) \approx \lambda_0^2$, so we arrive at $\Delta x\Delta\lambda/\lambda_0^2 \geq 1$.

De Broglie's formula relates the group velocity of the wave packet v_g to the wavelength and mass m of the particle $v_g = \mathrm{h}/m\lambda_0$. The range of velocities that determine the spreading of the packet is given by $\Delta v_g = h\Delta\lambda/m\lambda_0^2$

We use the definition of the momentum $\Delta \mathrm{p} = m\Delta v_g$ and $\Delta v_g = h\Delta\lambda/m\lambda_0^2$ to obtain Heisenberg's uncertainty principle $\Delta x\Delta \mathrm{p} \geq h$. Another form of the uncertainty principle relates to the standard deviation of the position σ_x and the standard deviation of momentum σ_p. This form is expressed as $\sigma_x\sigma_p \geq \hbar/2$.

Any usage of the terms "position" and "velocity" that exceeds the precision provided by $\sigma_x\sigma_p \geq \hbar/2$ is equally meaningless to the use of undefined words. This means that if we want to apply classical concepts such as position and velocity to a quantum system, we must acknowledge that there is a limit to how precisely these variables can be defined in the quantum context. As one variable becomes more particle-like, the other necessarily becomes less so.

Let us take a numerical example to understand how the uncertainty relation works. Suppose we can determine the position of an electron with an uncertainty of 10 nm. Using the uncertainty principle, we can find the minimum uncertainty in the electron's velocity as $m\Delta v \geq h/\Delta x$. The mass of an electron is $m_e = 9.10 \times 10^{-31}$ kg. Therefore, the minimum uncertainty in the velocity of the electron is $\Delta v \geq 72.74 \times 10^3$ m/s.

As discussed in Section 1.6.4.1, the uncertainty principle implies that no quantum state can simultaneously be a position and a momentum eigenstate. In other words, the wave function for position and momentum cannot be identical. However, this is not the case when the observables are compatible or commutable.

1.6.4.1 Uncertainty Principle and Fourier Transform

The Fourier transform is a powerful mathematical tool that finds applications in diverse fields of science and engineering. It represents a method for reconstructing a function by combining sine waves of different frequencies with appropriate amplitude and phase.

The uncertainty principle can be expressed using the language of the Fourier transform. A wave function can be represented either as a position-space wave function $\psi(x)$ or a momentum-space wave function $\phi(p)$.[2]

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{\frac{ipx}{\hbar}} \phi(p)\, dp \tag{1.26}$$

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{-\frac{ipx}{\hbar}} \psi(x)\, dx \tag{1.27}$$

It is clear that (1.26) and (1.27) (apart from the factors of $\hbar$) constitute a Fourier transform pair between the "position-space" and momentum space wave

2. This is analogous to the familiar time-domain and frequency-domain representations of a signal.

functions.[3] The Fourier transform bridges these two representations, allowing us to transform a wave function from one space to another.

The scaling property of the Fourier transform $\psi(ax) \overset{FT}{\leftrightarrow} (1/|a|)\phi(p/a)$ shows that if we scale the ψ wave function by the a factor ($a > 1$ corresponding to compression and $a < 1$ corresponding to the expansion of the wave function), then its Fourier transform ϕ is scaled by the $1/a$ factor. (See Figure 1.9.) Expansion (compression) in the position domain corresponds to compression (expansion) in the momentum domain. This behavior is related to the uncertainty principle, whereby the more compressed and less uncertain the position wave function becomes, the more extended and uncertain the momentum wave function becomes.

The only function with the same Fourier transform pair is the Gaussian wave function, which has the minimum uncertainty.

1.6.4.2 Uncertainty Principle and Commutation Relations

The uncertainty principle can also be expressed using a mathematical operator known as the commutator. The commutator of two operators $\hat{A}$ and $\hat{B}$ is defined as follows $[\hat{A},\hat{B}] \equiv \hat{A}\hat{B} - \hat{B}\hat{A}$.

An operator can be viewed as a matrix, and matrix multiplication is generally noncommutative. Therefore, if we have two operators, $\hat{A}$ and $\hat{B}$, in general, $AB \neq BA$. The uncertainty principle can be expressed in terms of commutators as follows: For two physical quantities to be concurrently observable, their operator representations must commute (that is, $[\hat{A},\hat{B}] = 0$). In other words, the order of measurement does not matter, and the result of the second measurement does

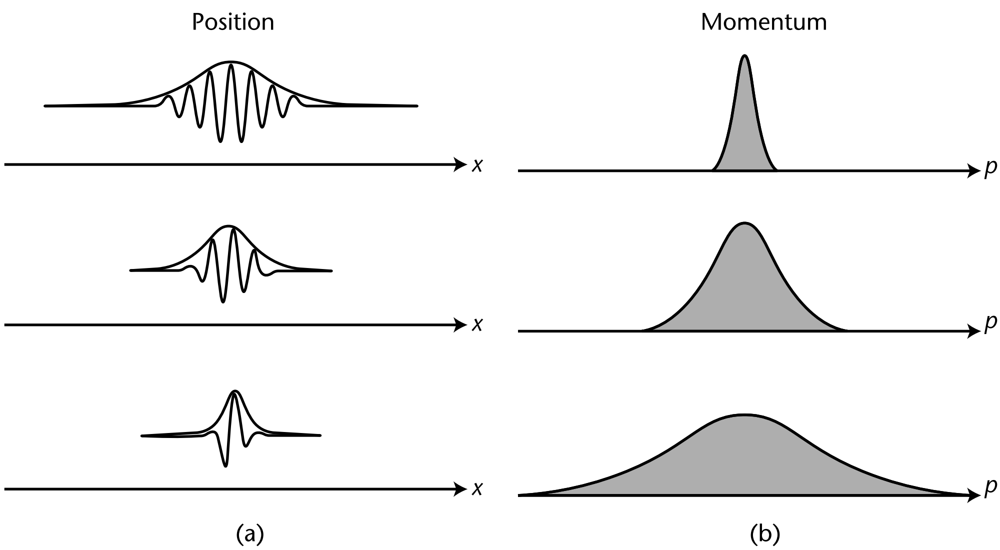


Figure 1.9 (a) The position domain (x-domain) wave function and (b) the corresponding Fourier transform in the momentum domain (p-domain). The scaling property of the Fourier transform shows that an expansion of the wavefunction in the x-domain results in the compression in the p-domain.

3. The probability of obtaining momentum in the range dp is $|\Phi(p, t)|^2 dp$.